tvet kigali
COMPETENCY-BASED ASSESSMENT
[FORMATIVE ASSESSMENT]

Program: SOFTWARE DEVELOPMENT | **Level:** 5 | **Total Marks:** 40
Module: Machine Learning Application | **Date:** 2026-05-27 | **Time:** 3 Hours

Name: ___________________________ Index No: ______________ Class: ________

SECTION A: TRUE / FALSE

Answer True (T) or False (F).

1. Supervised Learning replaces the need for any security when developing Machine Learning Model. **(1 marks)**

   Answer: **True** **False**

2. Logistic Regression must be configured before you can Machine Learning Model. **(1 marks)**

   Answer: **True** **False**

3. Support Vector Machine is an essential component when developing Machine Learning Model. **(1 marks)**

   Answer: **True** **False**

4. You can Machine Learning Model without ever using K-Means clustering. **(1 marks)**

   Answer: **True** **False**

SECTION B: MULTIPLE CHOICE

Choose ONE correct answer and circle the letter.

5. What is the main purpose of Reinforcement Learning in Machine Learning Application? **(1 marks)**

   A. Reinforcement Learning
   B. Semi-structured Data
   C. Model deployment in machine learning
   D. Memory limitation
   Answer: _______

6. What role does Test dataset play when developing Machine Learning Model? **(1 marks)**

   A. K-Nearest Neighbors
   B. Test dataset
   C. Neural Network
   D. Logistic Regression
   Answer: _______

7. Which statement best explains how Cross-validation works? **(1 marks)**

   A. Hyperparameter tuning
   B. Anaconda
   C. Cross-validation
   D. scikit-learn library
   Answer: _______

8. What is the correct way to implement Linear Regression? **(1 marks)**

A. Data preparation
B. Machine Learning
C. Data Gathering
D. Linear Regression
Answer: _______

9. How would you correctly use Random Forest when developing Machine Learning Model? **(1 marks)**

A. Hyperparameter tuning
B. Deep Q-Learning
C. Unsupervised Learning
D. Random Forest
Answer: _______

10. Which command or method is used to developing Machine Learning Model with Neural Network? **(1 marks)**

A. Logistic Regression
B. Neural Network
C. Supervised Learning
D. Model deployment in machine learning
Answer: _______

11. Which of the following correctly defines NumPy array? **(1 marks)**

A. NumPy array
B. ML Algorithm
C. Model deployment
D. Computation power
Answer: _______

SECTION C: MATCHING

Match each term in Column A with Column B.

12. Match each term in Column A with its correct description in Column B. **(2 marks)**

Column A (Terms)	Column B (Descriptions)
1. Supervised Learning	A. is the simplest machine learning model to understand in which
2. Linear regression	B. is the simplest machine learning model in which we try to predict one
3. The representation of linear regression	C. is a linear equation, which combines a set of
4. Logistic Regression	D. is a statistical model used to predict a binary outcome (i.e., 0 or 1,

Answers: 1.___ 2.___ 3.___ 4.___

SECTION D: OPEN-ENDED

Answer ALL questions. Write in the spaces provided.

13. Develop a plan for developing Machine Learning Model using Machine Learning. Include tools, steps, and expected results. **(6 marks)**

14. Evaluate the effectiveness of Unsupervised Learning when developing Machine Learning Model. Justify your answer with examples. **(3 marks)**

15. Explain how Training dataset works in the context of developing Machine Learning Model. **(3 marks)**

16. List and briefly describe the key components of F1 score used when developing Machine Learning Model. **(3 marks)**

17. Compare K-Nearest Neighbors with at least one alternative and explain which is better for developing Machine Learning Model. **(3 marks)**

18. Evaluate the effectiveness of Deep Learning when developing Machine Learning Model. Justify your answer with examples. **(3 marks)**

19. Define scikit-learn library and explain its role in developing Machine Learning Model. **(3 marks)**

20. Demonstrate step-by-step how to use Data preprocessing when developing Machine Learning Model. **(3 marks)**